

Ch - 3

Ch - 1.3.1

Types of Datasets →

Dataset - A dataset is a collection of related data used for analysis, data mining and machine learning.

A dataset consists of data objects and attributes.

- in detail
Later
- Data Object - An entity or individual record.
 - Attribute - A characteristic or property of a data object.

A. Document - Term Matrix →

A document term matrix (DTM) is used to represent text documents in a numerical or tabular form.

- Rows → represent documents
- Columns → represent terms or words
- Each value indicates presence or occurrence of a word in the document.

eg:

Document	Data	Mining	Machine
D ₁	2	1	0
D ₂	0	1	1

This representation helps in analyzing text data.

B. Transaction Database → It contains a set of transactions ↳ each transaction contains one or more items

eg:

Transaction ID	Item
T ₁	Bread, Milk
T ₂	Butter, Milk

This type of dataset is used in Association Rule Mining

↓

Bread → Milk

(It means customers who purchase bread may also purchase milk)

C. Time-Series Dataset - It contains data collected over a time.

eg:

Date	Temperature
Jan	20°C
Feb	24°C
March	28°C

Applications

- Stock price prediction
- Sales forecasting
- Weather forecasting

D. Spatial Dataset - It contains information related to geographical locations and spatial objects.

eg: Maps, GPS locations, Satellite images.

* Applications

- Traffic Analysis
- Location-Based services

E. Sequence Dataset → contains data where order of elements is important.

eg: Purchases := Laptop → mouse → keyboard.

* Applications

- DNA sequences
- web browsing history

F) Graph Datasets G) Streaming Datasets

STRUCTURED, UNSTRUCTURED & SEMI-STRUCTURED DATA →

A) Structured Data → organized in a fixed format usually rows & columns.

eg →

- Relational databases
- Student records
- Bank records etc.

* Characteristics

- Fixed Schema
- easy to store in tables
- easy to query using SQL
- Clearly defined attributes

eg:

ID	Name	Salary
101	Anjali	₹50000
102	Ram	₹10000

B) Semi-Structured Data - It does not follow a strict table format but contains tags, keys or markers that provide some organization.

eg: XML, JSON, HTML, Emails

```

{
  name: "Anjali",
  age: 20,
  course: "CSE"
}
  
```

* Characteristics

- flexible structure
- contains metadata or tags
- Schema can vary between records

C) Unstructured Data - It has no predefined format / schema.

eg → Images, Videos, Social media posts, Text documents, audio

* Characteristics

- Difficult to store in traditional tables
- * Requires preprocessing
- Often uses NLP, image processing or deep Learning Techniques

Characteristics of STRUCTURED DATA →

A) Dimensionality - It means number of attributes / features in a dataset.

eg:

ID	Name	Age	Course	Marks
----	------	-----	--------	-------

Here, the dataset has several attributes / dimensions.

Problem:

* Curse of Dimensionality - When no. of dimensions become very large, analysing data becomes difficult.

↑ No. of Dimensions ∝ Complexity ↑

* Solution: Feature selection / dimensionality reduction

B) ~~Sparsity~~ Sparsity - A dataset is sparse when many attributes values are zero/absent.

This is common in text data.

for eg: if there are 10,000 possible words, one document may contain only 100 words. Most values will ~~be~~ therefore be zero.

* Only presence counts - This means that in some applications, whether an item or feature is present can be more important than its exact value.

C) Resolution - Resolution refers to the level of detail or scale at which data is analyzed.

* Pattern depends on scale.

for eg. sales data may show diff patterns when analyzed

- Daily
- Monthly
- Yearly

A pattern visible at the monthly level may not be visible at daily level.

D) Distribution → describes how data values are spread

→ * Centrality → Represents the central or typical value of data.
eg: Mean, Median, Mode

→ * Dispersion → Dispersion shows how spread out data values are.
eg: Range, Variance, Standard deviation

Data Objects → A dataset is made up of data objects.

→ A data object represents an entity or individual records.

eg:

In a sales database:

- Customers
- orders
- Items

In a university DB

- Students
- Professors
- Courses

→ A data object can also be called:

- Sample
- Tuple
- Object
- Example
- Instance
- Data point

* Important Point → In a database:

Rows → data objects

Columns → Attributes

eg:

Student ID	Name	Age
101	Anjali	20
102	Rahul	21

Each row represents a student, so it is a data object.

Attributes → It is a data field that represents a characteristic or feature of a data object.

Attributes are also called:

- Dimensions
- Features
- Variables.

eg:

For customer: • Customer ID

• Name

• Address

• Age

These are all attributes
of the customer data object

* Types of Attributes

1) Nominal

2) Binary

3) Ordinal

4) Numeric

→ Interval - scaled

→ Ratio - scaled

1) Nominal Attribute - A nominal attribute represents categories, states or names of things

→ There is no meaningful order between the values.

eg. Hair Color: {Black, Brown, Grey, Red}

eg: (Red > Blue) has no meaning

∴ Nominal data only represent categories

2) Binary Attribute - A binary attribute is a nominal attribute that has only two possible states

→ usually represented as: 0 and 1

eg: True/False

Yes/No

A) Symmetric Binary - Both outcomes are equally ^{important}

B) Asymmetric Binary - The two outcomes are not equally important

3) Ordinal Attribute - It has meaningful order

An ordinal attribute has values with a meaningful order or ranking.

However, the exact difference between the successive values is not known.

eg: Small < medium < large.

There is an order, but we can't say how much larger is medium compared to small.

Hence, Order is meaningful, but the magnitude of difference is not known.

4) Numeric Attribute → Represent quantities.

They contain → Integer values
→ Real values.

eg: Age, Salary, Height, Temperature (°C, Fahrenheit)

A) Interval - Scaled Attribute

→ An interval scaled attribute is measured using a scale with equal-sized units.

→ The values have an order.

eg. $10^{\circ}, 20^{\circ}, 30^{\circ}$

The difference $30^{\circ} - 20^{\circ} = 10^{\circ}$
 $20^{\circ} - 10^{\circ} = 10^{\circ}$

* Interval data has no true zero points

B) Ratio Scaled Attribute : A ratio scaled attribute has an inherent or true zero point.

Because of true zeros, ratios are meaningful.

eg: Length, Counts, Temperature in Kelvin.

10kg, 5kg → 10kg is twice of 5kg.

This comparison is meaningful because weight has a true zero.

Interval	Ratio
→ No true zero	→ Has a True Zero
→ equal-sized units	→ equal sized units
→ Order exists	→ Order exists
→ Ratio are not meaningful	→ Ratio are meaningful
→ eg: Celsius, Fahrenheit	→ eg: lengths, counts, money

Based on Value →

A) Discrete Attributes → Has only a finite or countably infinite set of values.

eg: zip code, profession, etc.

→ Sometimes represented as integer values. eg: No. of Students: 20, 30

Binary attributes are special case of discrete attributes.

B) Continuous Attributes — It can have real numbers as attribute values.

eg: Height, Temperature, weight.

↓
160.2cm

↓
23.5°C

↓
40.2kg

→ They are typically represented as floating point variables

→ Real values can only be measured and represented using a finite number of digits practically

Discrete Attributes	Continuous
→ Usually countable	→ Usually measurable
→ Has finite values	→ Has real-number values
→ represented as integers	→ Represented as floating point values
eg: zipcode	eg: Height, weight

1.3.2

classmate

Why do we use Statistical description?

Statistical distribution helps us understand and summarize data.

Main purposes:

- Find central tendency → Where data is centered
- Find variation / dispersion → how spread out data is
- Detect outliers → usually high or low values.
- Understand the distribution / pattern of data.

Measures of Central Tendency →

Central tendency tells us the central or typical value of a dataset.

Main measures →
Mean
Median
Mode

I> A) Mean - It is the average of all values

* Sample Mean: If data is a sample of size n

$$\bar{x} = \frac{\sum x_i}{n}$$

where, n = sample size

x_i = individual value

eg: 2, 4, 6, 8, 10

$$\bar{x} = \frac{2+4+6+8+10}{5} = 6$$

* Population Mean: When we are dealing with entire population

$$\mu = \frac{\sum x_i}{N}$$

N = population size

μ = population mean

→ Mean is algebraic measure

B) Weighted Mean - When different values have different importance / weights.

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

x_i = value, w_i = weight.

C) Trimmed Mean - It is calculated by removing some extreme values from both ends and then calculating the mean.

• Why?

To reduce the effect of outliers

II. > Median - It is the middle value after arranging data in ascending / descending order.

* For Odd values $\rightarrow$ middle value

$$\left(\frac{n+1}{2} \right)^{\text{th}} \text{ observation}$$

eg: 10, 20, 30, 40, 50 $\left(\frac{5+1}{2} \right) = 3^{\text{rd}} = 30$

* For Even values $\rightarrow$

$$\frac{\left(\frac{n}{2} \right)^{\text{th}} + \left(\frac{n+1}{2} \right)^{\text{th}}}{2}$$

eg: 10, 20, 30, 40, 50, 60

$$= \frac{\left(\frac{6}{2} \right) + \left(\frac{6+1}{2} \right)}{2} = \frac{30+40}{2} = 35$$

* Median for Grouped Data $\rightarrow$ We estimate the median using ~~interf~~ interpolation.

$$\text{Median} = L_1 + \left(\frac{\frac{n}{2} - (\Sigma \text{freq})^L}{\text{freq}_{\text{median}}} \right) \times h$$

$\rightarrow$ class interval

L = lower boundary of median class
 n = no of observations = Σf

Σfreq^L = cumulative freq before median class

III> Mode - Most frequently occurring value.

eg: 2, 3, 3, 4, 5, 3, 6

Mode = 3

Types → unimodal → one mode
 → Bimodal → two modes
 → Trimodal → three modes

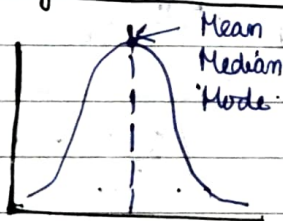
empirical Relationship -

$$\text{Mode} \approx 3(\text{Median}) - 2(\text{Mean})$$

OR

$$\text{Mean} - \text{Mode} \approx 3(\text{Mean} - \text{Median})$$

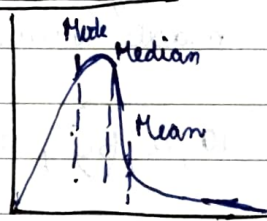
Symmetric Vs Skewed Data -



Symmetric

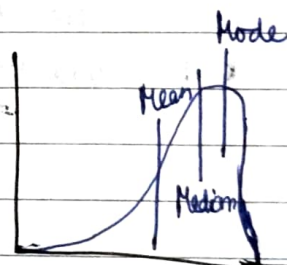


- Mean = Median = Mode
- The data is balanced around the center.



Positively (Right Skewed)
Skewed

- Mode < Median < Mean
- Tail extends towards the right.



Negatively (Left Skewed)
Skewed

- Mean < Median < Mode
- Tail extends towards the left.

Measures of Dispersion - Dispersion tells us how much data is spread out.

Important Measures

- Range
- Quartiles
- IQR
- Variance
- Standard Deviation
- Outliers

A) Range - Simplest measure of dispersion

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

• Example -

Data: 2, 5, 7, 10

$$\text{Range} = 10 - 2 = 8$$

• Limitation - Range depends only on the minimum and maximum values.

B) Quantiles - Quantiles divide sorted data into four parts.

Q_1 - First Quantile

→ 25th Percentile

→ 25% of observations below Q_1

Q_2 - Second Quantile

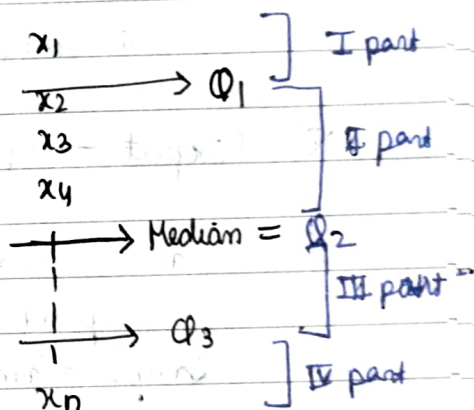
→ 50th Percentile

→ Same as Median

Q_3 - Third Quantile

→ 75th Percentile

→ 75% observations are below Q_3



C) Interquartile Range - IQR

IQR measures the spread of the middle 50% of data.

$$\text{IQR} = Q_3 - Q_1$$

eg: $Q_1 = 10, Q_3 = 30$

$$\text{IQR} = 30 - 10 = 20$$

D) Outliers - An outlier is a value that is ~~usually~~ ^{unusually} high or unusually low compared with the rest of the data.

Using IQR:

Lower Boundary $Q_1 - 1.5(\text{IQR})$

Upper Boundary $Q_3 + 1.5(\text{IQR})$

Values outside these boundaries are generally considered outliers

Example: Given $Q_1 = 10$, $Q_3 = 30$, ~~IQR~~

Sol. $\rightarrow IQR = Q_3 - Q_1$

$$= 30 - 10$$

$$= 20$$

$$\text{Lower Boundary} = Q_1 - 1.5(IQR)$$

$$= 10 - 1.5(20)$$

$$= -20$$

$$\text{Upper Boundary} = Q_3 + 1.5(IQR)$$

$$= 30 + 1.5(20)$$

$$= 60$$

Therefore, values below -20 or above 60 are outliers

~~E) Boxplot - A boxplot is a graphical representation of the~~

E) Variance - Variance measures how far data values are spread from the mean.

$$\text{Sample variance} - S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2 \right]$$

$$\text{Population variance} - \sigma^2 = \frac{\sum (x_i - \mu)^2}{N} = \frac{1}{N} \sum_{i=1}^n x_i^2 - \mu^2$$

$\rightarrow$ Variance is always non-negative

$\rightarrow$ Higher variance $\rightarrow$ data is more spread out

$\rightarrow$ Lower variance $\rightarrow$ data is more concentrated around the mean,

F) Standard Deviation $\rightarrow$ Standard deviation is the square root of variance

Sample $s = \sqrt{S^2}$

Population $\sigma = \sqrt{\sigma^2}$

Relation $\rightarrow$ Standard Deviation $= \sqrt{\text{Variance}}$

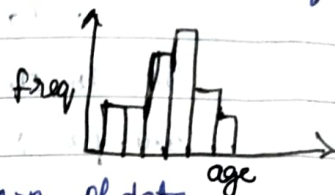
Graphical Displays of Data →

A) Histogram - A histogram displays the frequency distribution of numerical data.

• X-axis → values / intervals

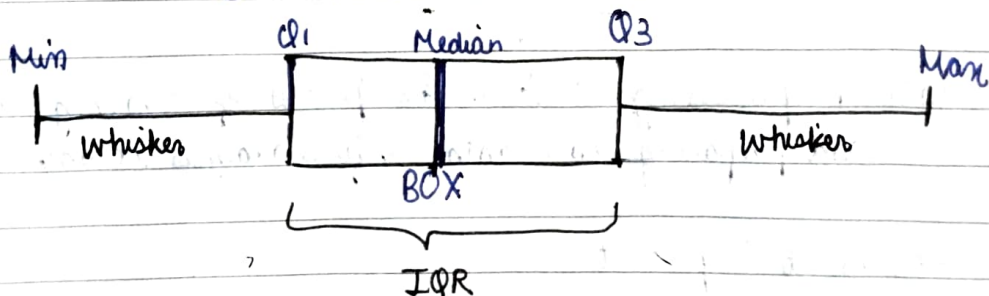
• Y-axis → frequencies

used to understand the distribution / pattern of data.



B) Boxplot - A boxplot is a graphical representation that shows the distribution and spread of data. It is based on the five-number summary:

- ① Minimum
- ② Q_1 (First Quartile)
- ③ Median (Q_2)
- ④ Q_3 (Third Quartile)
- ⑤ Maximum



It is a graphic display of five number summary.

C) Quantile Plot - A quantile plot pairs each value x_i with f_i , where approximately: $100 f_i \%$ of the data is less than or equal to x_i .
Purpose - Shows the distribution of data using quantiles.

D) Q-Q Plot :- Quantile-Quantile Plot compares the quantiles of two distributions.

→ Used to determine whether two distributions have similar pattern

→ One distribution's quantiles are plotted against the quantiles of another distribution.

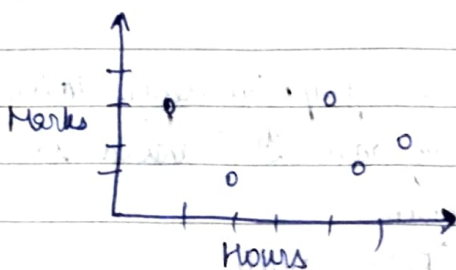
E) Scatter Plot → A scatter plot represents each pair of values as a point on a plane.

- X axis → one variable
- Y axis → another variable

Used to study the relationship / correlation between 2 variables.

Example →

Hours studies vs marks



1.3.3

Data Preprocessing, Visualization

Data Preprocessing → It is the process of cleaning, transforming and preparing raw data before analysis or machine learning.

Why is it important?

Raw data may contain

- Missing values
- Noise
- Inconsistent formats
- Duplicate data
- Different scales

Therefore, preprocessing improves the quality, accuracy, and efficiency of data analysis.

* Main Pre-processing Techniques →

- 1) Data Cleaning → handles missing, noisy and inconsistent data.
- 2) Data Normalization → brings values to a common scale.
- 3) Data transformation → converts data into a suitable format.
- 4) Data reduction → reduces the size / complexity of data.

Data Visualization → Data visualization is the process of representing data in a graphical or visual form, such as charts, graphs, maps, diagrams etc., so that data can be understood, analyzed and interpreted easily.

eg: Bar chart, boxplot, scatter plot, histogram, pie chart etc.

* Why Data Visualization is needed?

- 1) Easy understanding of data
- 2) Identifies patterns and trends
- 3) Gain insight into data
- 4) Identify relationships between data
- 5) Detect irregularities and anomalies
- 6) Support decision-making
- 7) Provide visual evidence of computer-generated results.

* Categorization of Visualization Methods →

Geometric Projection

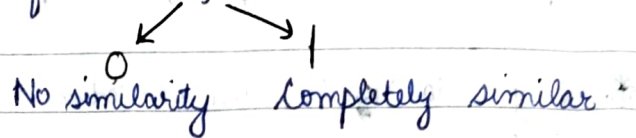
- 1) ~~Pixel - Oriented~~ Visualization - Data is represented using ^{geometric} objects / projections such as points, lines, etc. eg: Scatter Plots
- 2) Pixel - Oriented Visualization - Each data value is represented using a pixel.
- 3) Icon - Based Visualization - Data values are represented using icons or graphical symbols.
- 4) Hierarchical Visualization - Data is represented according to its hierarchical structure.
ex: Tree Structure
- 5) Visualization of complex data and relationships Used to visualize complex relationships between data objects.
eg: Networks / graphs

Measuring Data Similarity and Dissimilarity →

A) Similarity :-

It is a numerical measure of how alike two data objects are.

- Higher similarity → objects are more alike.
- Often ranges from: $[0, 1]$



Eg: If, $\text{Similarity}_{\text{score}}(A, B) = 0.9 \Rightarrow A \text{ and } B \text{ are highly similar.}$

B) Dissimilarity :-

It is a numerical measure of how different two data objects are.

- $\text{Similarity} \propto \frac{1}{\text{Dissimilarity}}$

- Lower dissimilarity → objects are more alike.
- Minimum is often 0.
- Maximum depends on the measure.

Eg: If, $\text{Distance}(A, B) = 2$ means A and B are relatively close according to that distance measure.

C) Proximity :- It is a general term referring to either:

- Similarity
- or
- Dissimilarity

$\therefore \text{Proximity} = \text{Similarity} \text{ or } \text{Dissimilarity}$

* Similarity $\uparrow \Rightarrow$ Alike-ness $\uparrow$
 Dissimilarity $\downarrow \Rightarrow$ Alike-ness $\uparrow$

Data Matrix VS Dissimilarity Matrix →

(A) Data Matrix → A data matrix represents data in tabular / matrix form.

→ It contains the actual attributes values of objects.

* If there are:

n → data objects / points

p → attributes or features

then matrix → $n \times p$

* Representation:

$$\begin{bmatrix} x_{11} & \dots & x_{1f} & \dots & x_{1p} \\ \vdots & & \vdots & & \vdots \\ x_{i1} & \dots & x_{if} & \dots & x_{ip} \\ \vdots & & \vdots & & \vdots \\ x_{n1} & \dots & x_{nf} & \dots & x_{np} \end{bmatrix}$$

* It contains two modes → Objects & Attributes

* Example:

Point	Attribute 1	Att. 2
x_1	1	2
x_2	2	5
x_3	3	0
x_4	4	5

fig: Sample data points with two attributes used for distance computation

(B) Dissimilarity Matrix → A dissimilarity matrix stores the distance / dissimilarity between pairs of objects.

• Matrix size → $n \times n$

• It is a triangular matrix because: $d(A, B) = d(B, A)$

so, we don't need to store both values.

* Representation

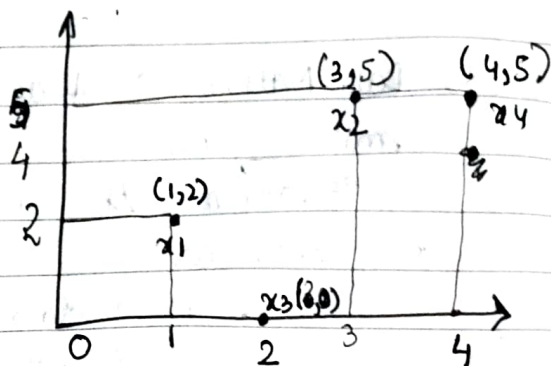
$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & \ddots & \\ d(n,1) & d(n,2) & d(n,3) & \dots & 0 \end{bmatrix}$$

* One Mode only

Example: Dissimilarity matrix with Euclidean distance.

Sol >

	x_1	x_2	x_3	x_4
x_1	0			
x_2	3.61	0		
x_3	4.24	5.1	0	
x_4	4.24	1	5.39	0



Distance b/w x_1 and x_2

$$x_1 = (1, 2), x_2 = (3, 5)$$

$$\begin{aligned} d(x_1, x_2) &= \sqrt{(1-3)^2 + (2-5)^2} \\ &= \sqrt{4+9} \\ &= \sqrt{13} = 3.61 \end{aligned}$$

Euclidean Distance

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

$$x = (x_1, x_2)$$

$$y = (y_1, y_2)$$

$$\begin{aligned} d(x_2, x_3) &= \sqrt{(3-2)^2 + (5-0)^2} \\ &= \sqrt{1+25} \\ &= \sqrt{26} \\ &= 5.1 \end{aligned}$$

$$\begin{aligned} d(x_4, x_3) &= \sqrt{(4-2)^2 + (5-0)^2} \\ &= \sqrt{4+25} \\ &= \sqrt{29} \\ &= 5.39 \end{aligned}$$

$$\begin{aligned} d(x_1, x_4) &= \sqrt{(1-4)^2 + (2-5)^2} \\ &= \sqrt{9+9} \\ &= \sqrt{18} = 4.24 \end{aligned}$$

$$\begin{aligned} d(x_2, x_4) &= \sqrt{(3-4)^2 + (5-5)^2} \\ &= \sqrt{1} \\ &= 1 \end{aligned}$$

Important Points →

- Data matrix → n data pts + p dimensions
- Dissimilarity matrix → n data pts + distance
- Data matrix → 2 modes
- Dissimilarity matrix → 1 mode

Properties of Data Matrix →

- ① Rows represent objects / records —
 - each row represents data instance, object, or tuple
 - eg: One row = One student
- ② Columns represent attributes / features —
 - each column represents a particular attribute / feature
 - eg: Marks, Age, CGPA
- ③ Structured format — Data is organized in tabular form.
 - Makes data easy to store, access / analyze.
- ④ Homogeneous column values —
 - Values within the same column usually have same datatype or domain.
 - eg: Age column contains numerical values.
- ⑤ Supports Preprocessing —
Data can be cleaned, normalized, transformed and reduced
- ⑥ Suitable for Machine Learning —
Most data mining and ML algos. use matrix-formatted data as input.